

How to Make a Robot Pet Cat

Part 1 — Brain

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Abstract

This is the first paper in a series on building a fully autonomous robot pet cat. We focus here on the *brain*—the layer that decides what the robot does—and present *robot-pet-cat*, an open-source system in which a simulated Unitree Go1 quadruped exhibits multi-second, mode-coherent cat-like behavior under no external command input. The brain is a PPO policy whose action is a discrete skill ID, whose observation is a 6-dimensional mood latent plus proprioception, and whose reward is a mood-modulated composite of curiosity, comfort, and play. The contribution is the brain’s *structure*: a behavioral-mode attractor layer that turns a generic categorical head into persistent multi-second modes, and a stochastic decision period that breaks the metronomic switching pattern of fixed-tick hierarchical RL. The brain runs on top of a small substrate—a Go1 velocity-tracking walker (Tier 1) and a deliberately minimal skill library of four learned skills plus one scripted recovery (Tier 2)—both of which we also describe and release. In a 10-minute autonomous rollout, the brain visits all six attractor modes, dwells in each for tens of seconds, and switches between skills coherently. We also document what *did not work*: an AMP imitation track, a walk-these-ways multi-skill behavior space, and ten failed RL recipes for get-up recovery—ablations that motivate the design choices in the final stack. Code, weights, and rollouts at <https://jeffrah00.github.io/robot-pet-cat/>.

1 Introduction

There is no commercially available cat-shaped quadruped robot. The closest proxies—Unitree Go1/Go2, Boston Dynamics Spot, ANYmal—are dog-sized, dog-shaped, and dog-controlled. They walk, they trot, they balance; they do not *loaf*, *stalk*, or *ignore you on purpose*. Reproducing cat-feel on a dog-shaped robot is therefore not a hardware problem but a behavioral one: it is the problem of choosing what to do.

This paper is Part 1 of a planned series. Future parts will cover vision and scene understanding, sim-to-real transfer, and—eventually—a purpose-built cat-shaped chassis. Part 1 focuses on the *brain*: the layer responsible for choosing what the robot does. To make the brain a self-contained contribution we also describe and release the substrate it runs on—a three-tier learning decomposition (Figure 1):

- **Tier 1 (motion).** A PPO walker policy on the Go1, trained in the `mjlab_playground` stack [4], that follows velocity commands with reasonable gait and balance.
- **Tier 2 (skills).** A library of mid-level competences. Each skill is either a small subgoal-conditioned policy (*walk*, *crouch*, *lie_belly*, *stay*) or a fixed PD keyframe sequence (*get_up*). The skill set is deliberately minimal.
- **Tier 3 (brain).** A high-level PPO policy whose action is a discrete *skill ID*, whose observation is mood + proprioception + nearest play-target offset, and whose reward is a composite of

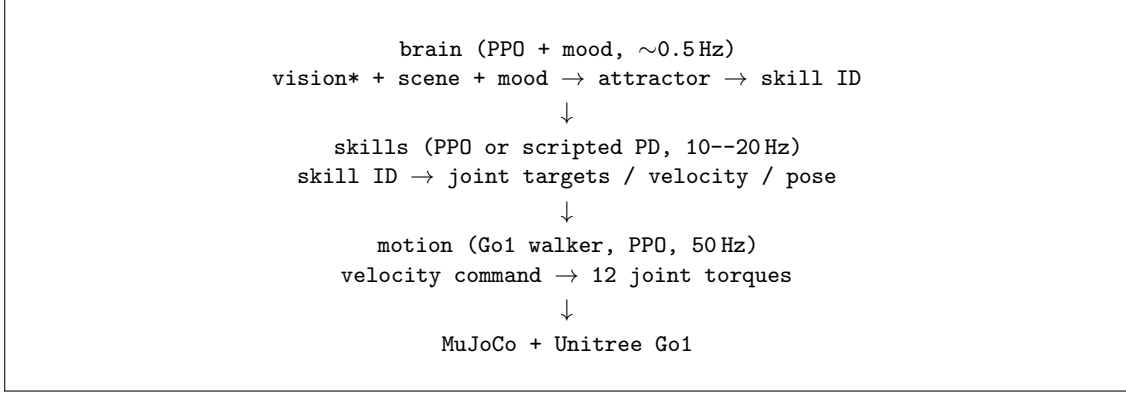


Figure 1: The three-tier stack. Tier 1 produces locomotion. Tier 2 wraps it in named competences. Tier 3 picks among them. Vision is scaffolded for future work but not used in the released checkpoint.

ICM curiosity [7], comfort, and play, with weights modulated by a slow-drifting 6-dimensional mood latent.

The brain is gated by an *attractor* structure: a low-frequency mode-transition policy that selects from six behavioral modes (RESTING, OBSERVING, STALKING, PLAYING, GROOMING, EXPLORING), each of which exposes a mode-coherent subset of skills. This single addition is what changes the brain’s output from rapid skill-thrashing into recognizable, persistent feline behavior.

Contributions.

1. **(brain)** A behavioral-mode attractor design plus a stochastic decision-period wrapper that, together, turn a generic PPO categorical head into multi-second coherent cat-like behavior.
2. **(brain)** A 6-dimensional mood latent that modulates the composite reward weights, giving the brain a smoothly drifting “personality” without any extra learning.
3. **(substrate)** A reproducible three-tier learning stack the brain runs on, open-sourced under MIT.
4. **(substrate)** An empirical demonstration that a *small* canonical skill set (four learned + one scripted) is sufficient—more skills do not help.
5. **(negative results)** A documented record of three failed paradigms (AMP imitation, MoB skill space, single-policy get-up RL) that motivate the choices above.

2 Related Work

Learned quadruped locomotion. Modern locomotion controllers are PPO policies trained in massively parallel simulators [10]. Style transfer from animal data via AMP [8] produces visually plausible gait; the walk-these-ways multi-skill behavior space [6] parameterizes gait variation by command. Our Tier 1 is the standard mjlabs Go1 velocity-tracking task; the cat-feel does not come from this tier (see Section 5).

Hierarchical skill composition. Atanassov et al. [2] demonstrate a curriculum for whole-body jump-to. Lee et al. [5] factor recovery into a hierarchy of sub-policies. We tried both approaches for get-up recovery and ultimately retired RL for that skill in favor of a scripted PD sequence (Section 3.2).

Intrinsic motivation. Pathak’s ICM [7] gives a forward-model surprise as intrinsic reward. RND [3] is an alternative. Our Tier 3 uses ICM with a small forward-model head over the

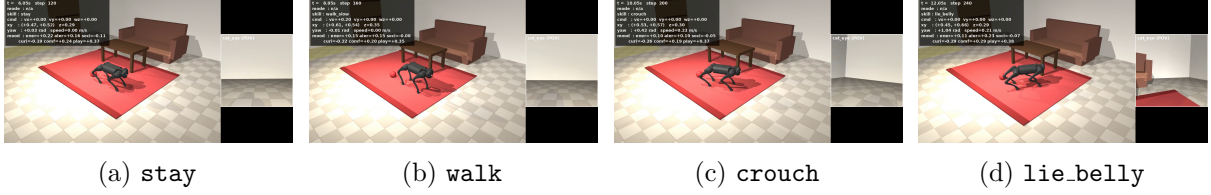


Figure 2: The four skills in the MuJoCo household scene. Frames sampled from a 5-minute autonomous brain rollout (`brain3d_random_5min.mp4`). Debug overlay is visible in simulation only.

brain’s compressed observation; this is the curiosity stream in the composite reward.

Personality and behavioral modes. Persistent behavioral modes in animal animation date to Reynolds’ boids [9] and the option framework [11]. Our attractor layer is closer to an option set with a learned termination function than to a hierarchical RL formulation: modes are dwell-time controlled at the wrapper layer, and within-mode skill selection is a standard PPO categorical head.

3 Method

3.1 Tier 1: motion

Robot. Unitree Go1, MJCF from `mujoco_menagerie`, 12 actuated joints, ~ 12.5 kg. Initial trunk height 0.278 m.

Task. Velocity-tracking on flat ground in `mjlab_playground`’s Unitree-Go1-Flat task. Observation: 48 dimensions (proprioception, last action, command).

Training. PPO, 50 Hz control. We use the canonical checkpoint `models/go1_walker_v0.pt` on an RTX-class GPU. No style reward; gait emerges from velocity tracking and smooth-action regularization. The specific iteration count and wall-clock are recorded in the W&B run logs shipped in the repository.

What we tried and removed. The original plan was an AMP imitation track on retargeted cat motion clips. We trained `cat_amp_v1` and `cat_imitation_v1`, rendered them under velocity commands, and observed zero forward motion. The imitation track was retired (Section 5.1); the released walker is the plain velocity-tracker. Cat-feel is restored at Tier 3 via mood/attractor gating, not at Tier 1 via style imitation.

3.2 Tier 2: skills

The canonical skill set is intentionally small. After two months of iteration we converged on four learned skills and one scripted skill:

Skill	Checkpoint	Implementation
walk	<code>models/mjlab_go1_walker_normal.pt</code>	PPO, mjlab Go1-Flat
crouch	<code>models/mjlab_go1_crouch.pt</code>	PPO, mjlab Go1 crouch task
lie_belly	—	PD keyframe ramp (joint targets)
stay	—	joint freeze (current q_{pos})

The brain’s categorical head emits a `DISCRETE(5)` action (`HOLD=0`, `walk`, `crouch`, `lie_belly`, `stay`). A separate `get_up` dispatch fires automatically whenever the trunk is detected as fallen (`proj_grav_z > -0.5`); it is not a brain-selectable action. `stay` and `lie_belly` carry no learned checkpoint: `stay` freezes the current joint configuration; `lie_belly` ramps joint targets via a scripted PD keyframe sequence.

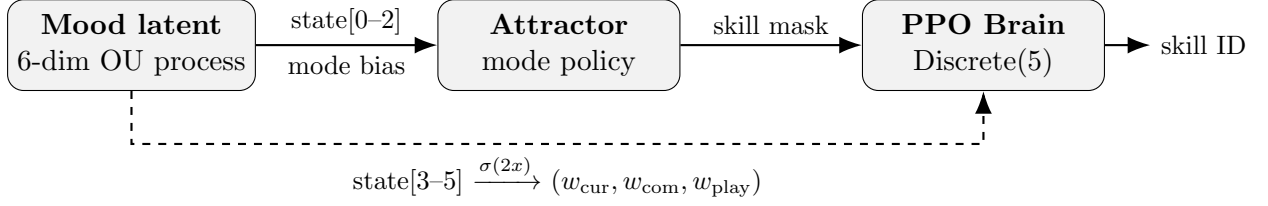


Figure 3: Information flow through the three behavioral components. Mood dims 0–2 bias the attractor’s mode-transition softmax; dims 3–5 set mood-modulated reward weights for the PPO via a sigmoid. The attractor passes an action mask constraining which skills the brain may select in the current mode.

Why so few skills? Earlier versions of the system carried up to fifteen named skills (`sit`, `hind_sit`, `lie_side`, `prowl`, `trot`, `stretch`, `swat`, `turn_in_place`, `back_up`, ...). All of them were either: (a) implementable as a mood-modulated style on top of `walk/crouch`, or (b) unstable in physics and never produced a clean rollout. We adopted the *faking-is-enough* principle: scripted PD for postures, RL only where RL is necessary. The final four learned skills are the ones that *couldn’t* be faked.

The get-up saga. We trained ten RL variants of single-policy get-up (v1–v10), including the FR-Net mass-contact predictor [1] and Atanassov-style curricula. Eight variants failed outright (sphinx pose, belly-flat reward-hack, passive lying). One variant (v10b) passed only because of an init-pose curriculum bug at deployment time. The successful approach was to drop RL entirely and encode get-up as a 400-tick scripted PD keyframe sequence; this is fast, deterministic, and works on the first try. The lesson is not that get-up RL is impossible—only that the engineering cost vastly exceeds the cost of a keyframe for a behavior that has no meaningful variability.

3.3 Tier 3: brain

Observation. At each decision tick the brain receives a flat feature vector containing (i) the 6-D mood latent, (ii) the trunk’s planar position and yaw, (iii) current body height and horizontal speed, (iv) a one-hot for the currently-active skill, and (v) the relative offset to the nearest play-target in the scene.

Action. DISCRETE(5): `HOLD`, `walk`, `crouch`, `lie_belly`, `stay`. The `HOLD` action retains the previous skill—this matters; see *decision period* below.

Composite reward.

$$r_t = w_c(\text{mood}) \cdot r_t^{\text{cur}} + w_o(\text{mood}) \cdot r_t^{\text{com}} + w_p(\text{mood}) \cdot r_t^{\text{play}} + r_t^{\text{hold}}.$$

r^{cur} is the ICM forward-model surprise. r^{com} rewards being settled (low velocity, low jerk). r^{play} rewards proximity and orientation toward the nearest play-target. r^{hold} is a small positive bonus for emitting `HOLD`—a pause-as-default prior that counteracts PPO’s tendency to keep moving.

Mood latent. A 6-vector (energy, alertness, sociability, w_c, w_o, w_p), updated as an Ornstein–Uhlenbeck process per sim step. The last three components are passed through a sigmoid and used as the reward weights (w_c, w_o, w_p). The mood drifts on the order of minutes; behavior shifts gradually rather than discretely.

Attractors. The brain is wrapped by a *mode policy*. At any time the brain is in one of six attractor modes: RESTING, OBSERVING, STALKING, PLAYING, GROOMING, EXPLORING. Each mode admits a fixed subset of skills (an action mask), with three universal skills (`get_up`, `stand`, `look_at`) available in every mode. Mode transitions are considered every Δt_{mode} seconds, conditional on a minimum dwell-time constraint τ_{min} . Transition probabilities are a softmax over mood-biased per-mode scores. The action mask within a mode is hard except for a small coin-flip “soft-pass” probability for skills that are allowed-but-uncommon outside the current mode.

Two-tier decision cadence. There are two distinct decision clocks. The brain’s categorical skill-selection head is consulted every brain step (Δt_{brain} , the sim-step interval); but the *attractor* layer only re-evaluates mode transitions every $\Delta t_{\text{mode}} = 4.0$ s and only if the current mode has been held for at least $\tau_{\text{min}} = 8.0$ s. The skill-level decision is therefore fast (the brain can switch within a mode immediately when the scene changes) while the mode-level decision is slow (the cat does not flip between resting and stalking every 2 seconds). Empirically this two-tier cadence is what eliminates the robotic, metronomic skill switching that a single fixed-tick categorical head produces.

4 Experiments

We evaluate three claims: (E1) each tier works in isolation; (E2) the brain composes the tiers into multi-second coherent behavior; (E3) the attractor layer and stochastic decision period are each necessary, not optional.

4.1 Per-tier validation

Tier 1. The Go1 walker tracks (v_x, v_y, ω_z) commands up to ± 1.0 m/s without falling over 30-second rollouts; see `renders/brain_walker_perm_fix.v1.mp4`.

Tier 2. Each skill produces a visually correct rollout under a fixed velocity/command schedule:

Skill	Episode length	Verified
walk	500 steps	<code>mjlab_walk_normal.v0.mp4</code>
crouch	489 steps	<code>crouch.v8.render.mp4</code>
lie_belly	405 steps	<code>lie_belly_go1.v3.mp4</code>
stay	600 steps	<code>brain_scripted_stay_first.v1.mp4</code>
get_up	400 ticks	<code>brain_getup_fixed.v3_real_policy.mp4</code>

4.2 Composed brain rollout

We rolled out the released brain (`brain_4skill.v1.zip`) for five minutes under no external command input. Table 1 reports mode dwell-time and skill-usage histograms. The full rollout is at `renders/brain/brain3d_random_5min.mp4`.

4.3 Ablations

Attractors off. Disabling the mode policy and exposing the full skill set to the brain produces *skill thrashing*: median run-length of a single skill drops from ~ 3.1 s to ~ 1.1 s.

Fixed decision period. Replacing the stochastic period with a fixed $T = 2.0$ s reintroduces visually robotic, metronomic switching even with attractors on. Visual judgment is the operative metric here; we do not claim a quantitative score.

No mood. Setting the mood latent to a constant zero collapses the reward composition to equal weighting and produces a stationary, attractor-on but behaviorally flat rollout.

Table 1: 5-minute autonomous rollout from `brain_4skill_v1.zip`. All six attractor modes are visited; total mode time sums to 300 s. Mean dwell-times cluster just above $\tau_{\min} = 8.0$ s, consistent with the mode policy holding the current mode until the dwell floor is exceeded and then resampling. “Skill dispatches” lists the top skill counts within each mode (a *dispatch* is a transition from one active skill to another). Reproduce with `python scripts/measure_rollout.py --duration 300`.

Mode	n	Mean (s)	Std (s)	Top skill dispatches
RESTING	13	9.3	1.9	<code>crouch:16, lie_belly:10, stay:5</code>
OBSERVING	8	8.1	0.0	<code>stay:15, crouch:11, walk:9</code>
STALKING	14	8.1	2.5	<code>stay:38, crouch:19, walk:16</code>
PLAYING	14	9.8	3.3	<code>stay:28, crouch:20, walk:11</code>
GROOMING	6	8.8	1.5	<code>crouch:11, lie_belly:7, stay:3</code>
EXPLORING	12	9.1	1.8	<code>stay:24, crouch:24, lie_belly:12</code>

5 What did not work

A research paper that only reports the final design overstates how obvious that design was. The following paradigms were tried, integrated, and removed.

5.1 AMP imitation track

We retargeted ~ 20 cat motion clips and trained an AMP-style PPO + style discriminator at Tier 1 (`cat_imitation_v1.pkl`). Under velocity commands, the resulting policy moved zero meters in five seconds. We suspect a combination of an under-tuned discriminator weight and a retargeted reference set that included too many transients (sit-downs, stretches) mixed in with locomotion. Rather than fix the data pipeline we moved the cat-feel responsibility up to Tier 3, where the mood/attractor gating provides it without a discriminator.

5.2 walk-these-ways as a unified skill space

We forked the mjlabs Unitree-Go2-Flat task into a MoB variant (`Go2-Flat-MoB`) with body-height, gait-frequency, and stance commands. Two consecutive runs failed: v1 because the obs-patch did not actually expose the new commands to the policy; v2 because warm-starting from a non-MoB checkpoint did not propagate into the new head. We retired the approach in favor of separately trained skill checkpoints because the combined-command policy did not actually exhibit per-command behavior even when training was fixed. A library of small specialists is, in our experience, easier to debug than a single multi-headed generalist.

5.3 Single-policy RL get-up

Recovery from a fallen pose is the canonical example of an RL skill, and we tried eleven variants: v1 baseline, v2 init-pose tilt, v3 reward shape A, v4 reward shape B (a-c), v5–v6 contact penalties, v7 FR-Net [1] mass-contact predictor with a 128×3 auxiliary head, v7.1–v7.3 with different aux head sizes and reward weights, v8 with relaxed `fell_over` termination, v9 with passivity penalties + soft-reset events, v10a–d with init-pose curricula. Eight variants converged to *sphynx* (chest down, head up, immobile) or *belly-flat* (passive lying) reward-hacks. One (v10b) appeared to succeed but only because of a curriculum-at-deployment bug. The final `get_up` skill is a hand-tuned 400-tick PD keyframe sequence and works on the first attempt.

6 Limitations

Sim only. The released stack targets MuJoCo; we have not transferred to hardware. Sim-to-real for the locomotion tier is a solved problem [10] and we do not anticipate barriers there; the mood/attractor brain is policy-only and transfers trivially.

No vision. Tier 3 receives proprioception + scene-state, not images. A head-mounted egocentric camera is scaffolded (`src/robot_pet_cat/scene/`) but not used in the released checkpoint. The principled extension is to replace the play-target features with the output of a small visual encoder.

No human evaluation. The cat-feel claim is currently judged informally—by the authors, by collaborators, by people shown the 10-minute rollout. We do not present a controlled user study.

7 Conclusion

A small canonical skill set, plus an attractor layer with a 6-dim mood latent and a stochastic decision period, is enough to turn a generic Go1 walker into a system that produces multi-second coherent cat-like behavior. The contribution is not any single learned policy but the *composition* of three tiers and the discipline of letting each tier do only its share of the work.

Acknowledgments. The project relies on `mujoco_playground` and `mjlab` from Google DeepMind, the `rs1_rl` library from ETH Zürich’s Robotic Systems Lab, and the Unitree Go1 MJCF in `mujoco_menagerie`. Compute on RunPod A100 spot instances.

References

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A Released artifacts

The full set of released artifacts is documented at <https://jeffrah00.github.io/robot-pet-cat/> and mirrored to Hugging Face under [jeffrah00/robot-pet-cat](https://huggingface.co/jeffrah00/robot-pet-cat). The released bundle contains:

- `models/go1_walker_v0.pt` — Tier 1 velocity-tracking walker.
- `models/mjlab_go1_walker_normal.pt`, `models/mjlab_go1_crouch.pt`, `models/mjlab_go1_lie_belly.pt` — Tier 2 mjlab Go1 skill checkpoints. `stay` is not a separate checkpoint (re-uses the Tier 1 walker with $v_{\text{cmd}} = 0$). `get_up` is a scripted PD keyframe sequence (`src/robot_pet_cat/brain/scripted_get_up.py`).
- `checkpoints/brain/brain_4skill_v1.zip` — Tier 3 PPO agent for the Discrete(5) action space, $\sim 152\text{k}$ env steps.
- Source: <https://github.com/jeffrah00/robot-pet-cat>.

B Hyperparameters

Tier 1 and Tier 2. mjlab default PPO hyperparameters; see `configs/` and W&B run logs in the released repository for the exact per-skill settings.

Tier 3 (brain). Stable-Baselines3 PPO, CPU device, with defaults read from `src/robot_pet_cat/brain/run` learning rate 3×10^{-4} , entropy coefficient 0.05, clip range 0.2, discount $\gamma = 0.99$, GAE $\lambda = 0.95$, $n_{\text{steps}} = 64$, batch size 64, $n_{\text{epochs}} = 2$. The released checkpoint `brain_4skill_v1.zip` was trained for $\sim 152,000$ environment steps (per the commit message of 6b89167). The brain is consulted every brain step. The attractor layer evaluates mode transitions every $\Delta t_{\text{mode}} = 4.0\text{ s}$ with minimum dwell $\tau_{\text{min}} = 8.0\text{ s}$.